Membership is Ownership: A Robust Model Watermark Detection with Quantile Regression

Anonymous Author(s)
Affiliation
Address
email

Abstract

Large-scale diffusion models are valuable intellectual property, yet can be stolen and adapted via downstream fine-tuning. Existing diffusion watermarking embeds explicit signatures into parameters or behavior, which may degrade utility and can be removed by post-hoc fine-tuning. We propose Membership Is Ownership, an ownership verification framework that treats a private member evidence set as a watermark and leverages the model's intrinsic memorization. Given a suspect model, we compute a multi-timestep *t-error* score by forward noising and single-step reconstruction, then aggregate with a lower quantile (Q25). To control false-positive rates, we fit a conditional Gaussian model in log-score space and obtain closed-form quantile thresholds at arbitrary target FPR. The protocol compares the suspect model against public baselines via hypothesis tests and effect sizes, producing statistical evidence rather than a single score. On CIFAR-10/100, STL-10, and CelebA, all datasets achieve $|d| > 16$ against public baselines; these separations persist after MMD fine-tuning (simulating post-theft adaptation), provided the adversary does not retrain from scratch.

1 Introduction

Large-scale generative diffusion models have become a central infrastructure for image synthesis, fueling a growing ecosystem of profitable downstream applications [Rombach et al., 2022]. The capability of these models is bought at considerable expense: producing a deployable checkpoint requires curated datasets, weeks of accelerator time, and substantial engineering investment, making the resulting weights valuable intellectual property (IP) for the institutions and individuals who train them. Yet these IP assets are exposed to a credible theft surface. An adversary who obtains a checkpoint through extraction attacks or insider leakage [Wang et al., 2025, Kim et al., 2024] can fine-tune it to a profitable domain, repackage it as a derivative product, and deploy it commercially without attribution. Establishing whether a suspect checkpoint in the adversary's possession originates from a particular owner's base model—*ownership verification*—is therefore a problem of practical economic and legal significance.

Existing approaches to diffusion-model IP protection address this problem along two complementary lines, each with structural limitations. *Training-time watermarking* embeds an explicit signature into model parameters or output behavior [Peng et al., 2023, Zhao et al., 2023, Fernandez et al., 2023, Wen et al., 2023, Wang et al., 2025, Ci et al., 2024, Li et al., 2025, Kim et al., 2024]: effective in controlled settings, but the embedded signature imposes a measurable utility cost (typically a non-trivial FID penalty) and can be progressively weakened by post-hoc fine-tuning, which an adversary will always have means to apply. *Membership inference attacks* (MIA) for diffusion models [Carlini et al., 2023, Duan et al., 2023, Kong et al., 2024, Matsumoto et al., 2023, Hu and Pang, 2023, Bertran et al., 2023] exploit the model's intrinsic memorization gap—reconstruction error on training samples is

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

systematically lower than on unseen samples—and avoid weight modification entirely. However, MIA is fundamentally a per-sample classifier: a low reconstruction error on a single image cannot anchor an ownership claim, both because individual samples may simply be easy to reconstruct and because modern diffusion models trained on overlapping public data exhibit shrinking memorization gaps that erode single-sample discriminability. Inference-time dataset inference [Dubiński et al., 2024] aggregates multiple MIA signals, but operates against open-ended public membership rather than a defender-controlled evidence set, leaving false-positive control implicit rather than calibrated.

The gap between these two lines suggests a third position. We propose **Membership is Ownership** (MiO), a verification framework built on a single principled observation: although the per-sample memorization signal is too weak to anchor ownership, the *population-level* memorization pattern over a defender-curated evidence set is both robust and statistically auditable. Concretely, the owner reserves a private subset $\mathcal{W}$ of the training data; this set is never disclosed and serves as a non-invasive watermark that leaves model parameters and output quality entirely intact. To extract a robust signal from a suspect model, we introduce a multi-timestep *t-error* score, computed by forward noising followed by single-step reconstruction across many timesteps, and aggregate it via the lower quartile (`Q25`) to suppress noise while retaining the timesteps where the memorization gap is widest. To convert per-sample scores into calibrated decisions at arbitrary false-positive budgets, we fit a conditional Gaussian model in log-score space using a bagged ensemble of quantile-regression networks, yielding closed-form thresholds at any target FPR without per-quantile retraining.

This formulation moves ownership verification out of single-sample classification and into a population-level hypothesis-testing problem. Given an owner model $\mathcal{M}_A$ trained on data containing $\mathcal{W}$, a suspect model $\mathcal{M}_B$ submitted for verification, and a set of public reference checkpoints representing the null hypothesis, the verifier's task is to decide between OWNED and INDEPENDENT under a controlled false-positive rate. Our threat model assumes white-box access to both the owner model and the suspect model—standard practice in IP litigation, regulatory audits, and contractual audit clauses—and permits the adversary to apply fine-tuning, distillation, or pruning to the stolen weights. A combinatorial argument bounds the probability that an adversary observing the full training set could guess $\mathcal{W}$ exactly at 10^{-5000} under the CIFAR-10 setting, rendering exhaustive evasion computationally infeasible. Full retraining from scratch lies outside the threat model, as it severs provenance entirely.

Verification proceeds via two statistical criteria that test different nulls. **C1 (consistency)** requires that the owner and suspect models exhibit indistinguishable score *distributions* on $\mathcal{W}$, operationalized through an unpaired Welch t-test ($p > 0.05$); failure indicates that the suspect model has no provenance relationship with the owner. **C2 (separation)** requires that the owner reconstructs $\mathcal{W}$ significantly better than every public reference, operationalized through a one-sample test on the paired difference $\delta_{\text{C2}}(x) = s_{\mathcal{M}_A}(x) - s_{\text{ref}}(x)$ with effect size $|d|_{\text{C2}}$; we require both statistical significance ($p < 10^{-6}$) and practical significance ($|d|_{\text{C2}} > 2.0$). The paired form applies uniformly: when owner and reference share base weights (as in PEFT settings), pairing cancels per-image variance and is essential for retaining power; when they do not, pairing remains valid with negligible loss. Ownership is declared verified only when both criteria hold jointly, ensuring that no single metric can yield a false positive: C1 confirms provenance, C2 establishes that the memorization signal is genuine rather than coincidental.

We validate the framework across two evaluation scenarios chosen for complementary epistemic purposes. The first uses DDIM models trained from scratch on CIFAR-10, CIFAR-100, STL-10, and CelebA, providing the controlled environment necessary for measuring the protocol's quantitative properties with precision. The second evaluates Stable Diffusion v1.4 under LoRA at ranks $r \in \{64, 256\}$ and full fine-tuning, reflecting the production-scale threat model where actual diffusion-model IP theft occurs. Across all four DDIM datasets, every public reference yields paired separation $|d|_{\text{C2}} > 16$, an order of magnitude above the verification threshold; the separation persists under MMD fine-tuning, SGD fine-tuning, additive Gaussian weight noise, and 2,000-step adversarial PEFT, simulating realistic post-theft adaptation. Across all three SD owner configurations and both adversary types (domain-shift and task-shift), the protocol returns a verified verdict in every case. In a controlled comparison against WDM [Peng et al., 2023], Zhao et al. [Zhao et al., 2023], CDI [Dubiński et al., 2024], and SleeperMark [Wang et al., 2025], MiO matches the post-theft robustness of training-time watermarking baselines while incurring zero training overhead, zero weight modification, and zero FID cost.

We summarize the contributions of this paper as follows.

- **Conceptual.** We reframe ownership verification from single-sample membership inference into a population-level hypothesis-testing problem, in which a private member evidence set held by the owner serves as a non-invasive watermark that leaves model parameters and output quality intact.
- **Methodological.** We combine the multi-timestep `Q25` t-error score with a Gaussian quantile regression in log-score space and a two-point verification protocol (consistency via unpaired Welch test; separation via paired Cohen's d). The framework yields closed-form FPR control at arbitrary significance levels without retraining.
- **Empirical.** We validate the framework on DDIM across four datasets and on Stable Diffusion v1.4 across three owner configurations. All DDIM-from-scratch settings achieve $|d|_{\mathrm{C2}} > 16$ against public references, with the verdict robust to MMD, SGD, weight-noise, and PEFT adversaries. Every SD owner configuration passes the verification protocol under both domain-shift and task-shift adversarial fine-tuning.
- **Comparative.** Against WDM, Zhao et al., CDI, and SleeperMark, MiO matches post-theft robustness while requiring no weight modification, no training overhead, and no FID cost—properties unique to operating against a private member evidence set rather than an embedded signature.

2 Related Work

Model Watermarking. Prior machine learning model watermarking methods typically embed an identifiable pattern into model behavior, most notably through trigger-based backdoors or carefully constructed inputs that induce a predetermined response [Adi et al., 2018, Zhang et al., 2018, Fernandez et al., 2023, Wen et al., 2023]. Such approaches provide a direct verification interface when the watermark remains intact, but they generally require explicit intervention during training and can be sensitive to subsequent model editing or "cleaning." *In contrast to embedding an external trigger, our setting treats a private member evidence set as the "watermark" and derives evidence from the model's intrinsic memorization behavior, avoiding modifications that introduce new behavioral artifacts.*

Membership Inference Attacks. Membership inference attacks (MIA) aim to decide whether an input was part of a model's training data [Shokri et al., 2017, Yeom et al., 2018, Salem et al., 2019]. For diffusion models, recent work instantiates MIA using reconstruction-based signals [Carlini et al., 2023, Duan et al., 2023, Kong et al., 2024], loss/likelihood-based criteria [Matsumoto et al., 2023, Hu and Pang, 2023], or scalable quantile-regression formulations [Bertran et al., 2023, Tang et al., 2024]. These methods are primarily evaluated as per-example decision rules (binary classification or scoring) and are typically reported via attack metrics. Ownership verification, however, requires comparative, population-level evidence: the ownership claim must be supported by aggregate separation on a designated evidence set, calibrated against reference models that represent the null hypothesis, and summarized via statistically interpretable tests rather than single-sample predictions. *Our framework uses MIA-style signals as the underlying measurements, but creates an auditable verification procedure with public references and formal decision criteria.*

Diffusion Model IP Protection. Diffusion-specific IP protection has developed along two complementary lines. *Training-time watermarking* embeds a recoverable signal during training; representatives for unconditional diffusion include WDM [Peng et al., 2023] and Zhao et al. [Zhao et al., 2023], while Stable Diffusion has attracted dedicated designs such as Tree-Ring [Wen et al., 2023] (a Fourier-domain noise pattern), SleeperMark [Wang et al., 2025] (a trigger-prompt-coupled, LoRA-resistant watermark), Stable Signature [Fernandez et al., 2023], RingID [Ci et al., 2024], GaussMarker [Li et al., 2025], and WOUAF [Kim et al., 2024]. *Inference-time dataset inference* is a complementary, training-free line: CDI [Dubiński et al., 2024] aggregates seven membership inference attacks through a meta-classifier to produce a dataset-level membership score. *Two framework-level distinctions separate our work from these prior approaches.* (i) *Training-time vs. inference-time:* the training-time methods above all require a modified training pipeline and incur measurable FID cost, whereas our framework attaches to an already-trained model with $\Delta\mathrm{FID}{=}0$ and zero training overhead.

(ii) *Open-ended vs. private-set verification:* CDI shares our inference-time, no-FID-penalty regime but operates against open-ended public-DDPM membership, whereas we leverage a private member evidence set held by the owner to enable calibrated population-level hypothesis testing under controlled false-positive budgets. Empirical comparison against WDM, Zhao, and CDI is given in Section 5.5 (Tables 6 and 5).

3 Problem Setup

3.1 Background and Problem Formulation

Diffusion Models. A diffusion model defines a forward noising process and a learned reverse process. Given a clean sample $x_0 \sim q(x_0)$, the forward process produces a noisy version at timestep $t \in \{1, \ldots, T\}$:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise schedule. A neural network ϵ_θ is trained to recover the added noise by minimizing:

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0,\, \epsilon,\, t}\big[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\big]. \tag{2}$$

At inference, new samples are drawn by iteratively denoising from $x_T \sim \mathcal{N}(0, I)$ using the learned ϵ_θ.

It has been empirically established that diffusion models memorize portions of their training data Carlini et al. [2023], Somepalli et al. [2023]: the reconstruction error on training examples is systematically lower than on unseen samples drawn from the same distribution. This memorization gap persists across architectures and dataset scales, and serves as the foundational signal that our framework exploits for ownership verification.

From Membership Inference to Ownership Verification. Membership inference attacks (MIAs) leverage the memorization gap to determine whether a single query sample belongs to the training set of a target model Duan et al. [2023], Kong et al. [2024]. We state this formally:

Definition 1 (Membership Inference Attack)**.** *Given a target model M and a query sample x, a membership inference attack is a binary classifier $\mathcal{A} : (M, x) \to \{0, 1\}$ that predicts whether $x \in \mathcal{D}_{\text{train}}$.*

However, low reconstruction error on a single sample does not constitute evidence of ownership, as the sample may simply be inherently easy to reconstruct for any model. Ownership verification demands a stronger, population-level claim: the suspect model must exhibit statistically significant memorization on a *designated evidence set* when compared against public baselines that share no training overlap with that set.

Definition 2 (Ownership Verification)**.** *Let $\mathcal{M}_A$ denote a base model trained by the owner, $\mathcal{M}_B$ a suspect model held by the adversary, and $\mathcal{W} = \{x_1, \ldots, x_{|\mathcal{W}|}\}$ a private evidence set held by the owner such that $\mathcal{W} \subset \mathcal{D}_{\text{train}}(\mathcal{M}_A)$. An ownership verification protocol is a hypothesis test $\mathcal{V} : (\mathcal{M}_B, \mathcal{W}, \{M_{ref}^{(i)}\}) \to \{\texttt{owned}, \texttt{independent}\}$ that determines whether $\mathcal{M}_B$ is derived from $\mathcal{M}_A$ by comparing the reconstruction behavior of $\mathcal{M}_B$ on $\mathcal{W}$ against a set of public baseline models $\{M_{ref}^{(i)}\}_{i=1}^{|\mathcal{R}|}$ at a controlled false-positive rate α.*

This formulation shifts the task from sample-level binary classification to a calibrated, population-level statistical test, which is the protocol we develop in the following sections.

3.2 Threat Model

We consider three principals in the ownership verification setting. The **owner** O trains a model $\mathcal{M}_A$ on $\mathcal{D}_{\text{train}}$, which contains a private watermark set $\mathcal{W} \subset \mathcal{D}_{\text{train}}$ known only to O. The **adversary** obtains $\mathcal{M}_A$ through theft or leakage, and produces a derivative model $\mathcal{M}_B$ by fine-tuning; $\mathcal{M}_B$ is the suspect model submitted for verification. The **verifier** V, given $\mathcal{M}_A$, $\mathcal{M}_B$, a set of public baseline models, and the private set $\mathcal{W}$, determines whether the ownership claim is valid.

We assume the adversary has white-box access to $\mathcal{M}_A$ and may possess in-domain data disjoint from $\mathcal{W}$. The adversary is aware that a verification protocol exists but cannot identify $\mathcal{W}$: even if $\mathcal{D}_{\text{train}}$ were leaked, selecting the exact $|\mathcal{W}|$-subset from N training samples is feasible with probability at most

$$P(\text{exact match}) = \frac{1}{\binom{N}{|\mathcal{W}|}} \leq \left(\frac{|\mathcal{W}|}{N}\right)^{|\mathcal{W}|}, \tag{3}$$

which is upper-bounded by 10^{-5000} under the CIFAR-10 setting ($N = 50{,}000$, $|\mathcal{W}| = 5{,}000$), making exhaustive guessing computationally infeasible. This is a counting argument and does not claim resistance to side-channel leakage of $\mathcal{W}$. The adversary is permitted to apply fine-tuning, distillation, and pruning to $\mathcal{M}_A$; full retraining from scratch falls outside our scope, as it severs provenance entirely.

Upon receiving a dispute, the verifier computes t-error scores on $\mathcal{W}$ for both the suspect model and public baselines, then evaluates the two-point criteria (Section 4.4), outputting VERIFIED or REJECTED together with the supporting statistical evidence (p-values and Cohen's d). The verifier is assumed to have white-box access to both models, consistent with standard practice in IP litigation, regulatory audits, and contractual audit clauses.

4 Method

The verification framework has four components: (1) a *t-error* score measuring reconstruction quality via forward-reverse diffusion, (2) multi-timestep aggregation for stable per-sample scores, (3) Gaussian quantile regression for FPR-controlled thresholds, and (4) a two-point statistical protocol comparing the suspect model against public baselines.

4.1 t-error Scoring

Diffusion models reconstruct their training data with lower error than unseen data [Carlini et al., 2023]. We exploit this *memorization gap* by measuring reconstruction quality at a given noise level t.

Specifically, given a clean sample x_0, we first corrupt it via the forward diffusion schedule:

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{4}$$

where x_t is the noised image at timestep $t \in \{1, \ldots, T\}$, ϵ is i.i.d. Gaussian noise, and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise coefficient under a cosine schedule [Dhariwal and Nichol, 2021]. The model then predicts the added noise $\hat{\epsilon}_\theta(x_t, t)$, from which we recover a single-step reconstruction:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon}_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{5}$$

We define the *t-error*, denoted $e_t(x_0)$, as the per-pixel mean squared error between the original and reconstructed images:

$$e_t(x_0) = \frac{\|x_0 - \hat{x}_0\|_2^2}{H \times W \times C} \tag{6}$$

where H, W, C denote the image height, width, and number of channels, respectively. The key property underlying our framework is the *memorization effect*: models trained on a particular dataset exhibit systematically lower reconstruction error on their training examples, i.e., $\mathbb{E}_{x \in \mathcal{D}_{\text{train}}}[e_t(x)] < \mathbb{E}_{x \notin \mathcal{D}_{\text{train}}}[e_t(x)]$. This persistent gap, which survives across architectures and dataset scales, provides the foundational signal for our ownership verification protocol.

For latent diffusion models that operate in VAE latent space (e.g., Stable Diffusion), the definitions above transfer directly: the t-error computation is moved from pixel space to latent space, while the rest of the framework remains unchanged. The latent-vs-pixel design choice and its empirical justification are in Section 5.6.

4.2 Multi-Timestep Aggregation

A single timestep provides a noisy estimate of reconstruction quality, as the magnitude of $e_t(x)$ varies substantially with the noise level t. To obtain a stable per-sample statistic, we aggregate t-errors over

Algorithm 1 t-error Computation

Require: Model $\mathcal{M}$, sample x_0, timesteps $\mathcal{T} = \{t_1, \ldots, t_K\}$, noise schedule $\{\bar{\alpha}_t\}$
Ensure: Aggregated score $s(x_0)$
  errors $\leftarrow []$
  for $t \in \mathcal{T}$ **do**
    $\epsilon \sim \mathcal{N}(0, I)$
    $x_t \leftarrow \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$
    $\hat{\epsilon} \leftarrow M_\theta(x_t, t)$
    $\hat{x}_0 \leftarrow (x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon})/\sqrt{\bar{\alpha}_t}$
    $e_t \leftarrow \|x_0 - \hat{x}_0\|_2^2/(H \times W \times C)$
    errors.append(e_t)
  end for
  return $s(x_0) \leftarrow Q_{0.25}(\texttt{errors})$

K uniformly sampled timesteps into a single score $s(x)$ via the 25th percentile:

$$s(x) = Q_{0.25}\big(\{e_{t_i}(x)\}_{i=1}^{K}\big) \tag{7}$$

We use K=50 by default. The choice of a lower percentile is deliberate: it captures the timesteps where the memorization gap is most pronounced while remaining robust to outlier timesteps that introduce high but uninformative error. Ablations in Section 5.6 confirm that the 25th percentile yields the strongest separation among common aggregation strategies, and that performance is stable for $K \geq 25$. The full scoring procedure is summarized in Algorithm 1.

4.3 Gaussian Quantile Regression

Rather than relying on fixed thresholds, we model the *conditional score distribution* for non-members and derive decision boundaries in closed form. Concretely, we apply a log transformation to the aggregated score and parameterize the result as a Gaussian with input-dependent mean $\mu(x)$ and standard deviation $\sigma(x)$:

$$y = \log(1 + s(x)) \sim \mathcal{N}\big(\mu(x),\, \sigma^2(x)\big) \tag{8}$$

where $\mu(x)$ and $\sigma(x)$ are jointly predicted by a neural network f_ψ. The network uses a ResNet-18 backbone to extract image features and concatenates them with a compact statistical summary $\phi(x) \in \mathbb{R}^3$. This summary comprises three quantities: the mean, standard deviation, and ℓ_2 norm of the t-error sequence. An MLP head maps the combined representation to $(\mu, \log\sigma)$. These auxiliary statistics provide a direct view of the t-error magnitude that the image features alone cannot capture, stabilizing the quantile predictions.

We train f_ψ by minimizing the Gaussian negative log-likelihood over N auxiliary non-member samples:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{(y_i - \mu_i)^2}{2\sigma_i^2} + \log\sigma_i\right] \tag{9}$$

where $y_i = \log(1 + s(x_i))$ is the observed log-score and (μ_i, σ_i) are the corresponding network predictions. Because the conditional distribution is Gaussian, the $(1-\alpha)$-quantile at any target false-positive rate α admits a closed-form expression:

$$\hat{q}_\tau(x) = \mu(x) + \sigma(x) \cdot \Phi^{-1}(\tau), \quad \tau = 1 - \alpha \tag{10}$$

with Φ^{-1} denoting the standard normal inverse CDF. This is a key practical advantage over pinball-loss quantile regression, which requires training a separate model for every desired quantile level; here, a single model supports arbitrary τ without retraining.

To translate the predicted quantile into a membership decision, we define the *margin* $m_\tau(x)$ as the gap between the predicted quantile and the observed log-score:

$$m_\tau(x) = \hat{q}_\tau(x) - \log(1 + s(x)) \tag{11}$$

A positive margin indicates that the sample's reconstruction error falls below what would be expected of a non-member, suggesting likely membership in the training set.

Finally, to reduce prediction variance, we wrap the above procedure in a bagging ensemble of $B{=}50$ models, each trained on a bootstrap sample drawn with replacement from 80% of the auxiliary data. At inference, we average the individual quantile predictions into a single ensemble estimate:

$$\hat{q}_\tau^{\text{ens}}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{q}_\tau^{(b)}(x) \tag{12}$$

where $\hat{q}_\tau^{(b)}(x)$ is the quantile from the b-th ensemble member. The ensemble margin $m_\tau^{\text{ens}}(x) = \hat{q}_\tau^{\text{ens}}(x) - \log(1 + s(x))$ serves as the final per-sample statistic; as before, positive values indicate likely membership. Ensembling is embarrassingly parallel and introduces negligible additional inference cost (Appendix C).

4.4 Two-Point Verification Criteria

Given the owner model $\mathcal{M}_A$, the suspect model $\mathcal{M}_B$, a set of public references $\{M_{\text{ref}}^{(i)}\}$, and the watermark set $\mathcal{W}$, we require both of the following criteria to be satisfied before declaring ownership verified. The two criteria test different nulls and use different statistics:

$$\delta_{\text{C1}}(x) = s_{\mathcal{M}_A}(x) - s_{\mathcal{M}_B}(x), \tag{13}$$
$$\delta_{\text{C2}}(x) = s_{\mathcal{M}_A}(x) - s_{\text{ref}}(x). \tag{14}$$

Criterion 1: Consistency. If $\mathcal{M}_B$ is indeed derived from $\mathcal{M}_A$, the two models should exhibit similar reconstruction *distributions* on $\mathcal{W}$. We operationalize this with a two-sample Welch t-test on the raw score sequences $\{s_{\mathcal{M}_A}(x)\}_{x\in\mathcal{W}}$ and $\{s_{\mathcal{M}_B}(x)\}_{x\in\mathcal{W}}$, requiring $p > 0.05$. The unpaired form is necessary here: when $\mathcal{M}_B$ is a benign fine-tune of $\mathcal{M}_A$, the per-image correlation between their scores is near unity, and a paired one-sample test on δ_{C1} would over-reject by detecting any systematic per-image drift induced by fine-tuning. Distributional equality is the right null for provenance; per-image identity is not.

Criterion 2: Separation. The owner model must reconstruct watermark samples significantly better than any public reference. We apply C2 to the paired difference δ_{C2} via a one-sample t-test against zero, with effect size

$$|d|_{\text{C2}} = \frac{|\,\text{mean}(\delta_{\text{C2}}(\mathcal{W}))\,|}{\text{std}(\delta_{\text{C2}}(\mathcal{W}))}, \tag{15}$$

and require both statistical significance ($p < 10^{-6}$, to control for multiple reference comparisons) and practical significance ($|d|_{\text{C2}} > 2.0$, corresponding to a "very large" effect size [Cohen, 1988]). The paired form applies regardless of whether owner and reference share base weights. When they do (as in the SD setting), pairing is essential because per-image variance shared via the base dominates the unpaired denominator and the unpaired effect size collapses. When they do not (as in the DDIM-from-scratch setting, where the per-image score correlation $\rho \in [0, 0.6]$), pairing remains valid; relative to an unpaired analysis, the paired effect size differs by a factor that depends on ρ and the variance ratio between owner and reference, but with $|\mathcal{W}| \geq 1{,}000$ and the effect sizes observed in our experiments the resulting margins remain orders of magnitude above the threshold (Section 5.2).

Ownership is declared **verified** only when both criteria are jointly satisfied. The conjunction ensures that no single metric can produce a false positive: consistency confirms provenance, while separation establishes that the memorization signal is genuine rather than coincidental. The complete verification protocol is formalized in Algorithm 2.

5 Experiments

5.1 Experimental Setup

5.1.1 Evaluation Scenarios

We evaluate the verification protocol in two distinct scenarios chosen for complementary purposes.

The first scenario uses DDIM models trained from scratch on four standard datasets. Training every component end-to-end gives us interpretable model architecture, controlled training data, and minimal

Algorithm 2 Ownership Verification Protocol

Require: Models M_A, M_B, references $\{M_{\text{ref}}^{(i)}\}_{i=1}^{|\mathcal{R}|}$, watermark set $\mathcal{W}$
Ensure: VERIFIED or REJECTED
  $S_A \leftarrow \{s_{M_A}(x) : x \in \mathcal{W}\}$; $S_B \leftarrow \{s_{M_B}(x) : x \in \mathcal{W}\}$
  C1 (consistency, unpaired): $p_1 \leftarrow \text{WelchTTest}(S_A, S_B)$
  if $p_1 < 0.05$ **then**
    return REJECTED
  end if
  C2 (separation, paired): for each reference $i \in \{1, \ldots, |\mathcal{R}|\}$:
    $S_{\text{ref}}^{(i)} \leftarrow \{s_{M_{\text{ref}}^{(i)}}(x) : x \in \mathcal{W}\}$
    $\delta_{\text{C2}}^{(i)} \leftarrow S_A - S_{\text{ref}}^{(i)}$ {element-wise on $\mathcal{W}$}
    $p_2^{(i)} \leftarrow \text{OneSampleTTest}\big(\delta_{\text{C2}}^{(i)}, \mu_0 = 0\big)$
    $|d|_{\text{C2}}^{(i)} \leftarrow \big|\,\text{mean}(\delta_{\text{C2}}^{(i)})\,\big| / \text{std}(\delta_{\text{C2}}^{(i)})$
  if $\exists i : p_2^{(i)} > 10^{-6}$ **or** $|d|_{\text{C2}}^{(i)} < 2.0$ **then**
    return REJECTED
  end if
  return VERIFIED

statistical noise—necessary conditions for measuring the protocol's quantitative properties with precision. This scenario also enables controlled comparison against existing baselines (Section 5.5), since the owner training pipeline matches what those baselines were designed for.

The second scenario evaluates Stable Diffusion v1.4 with parameter-efficient fine-tuning. The owner is a production-scale pretrained model adapted to a private dataset, with adversaries operating via PEFT on disjoint data—the threat surface where actual diffusion-model IP theft occurs. This scenario is necessary for external validity: a protocol evaluated only on DDIM trained from scratch leaves open whether it works on the kind of model that gets stolen. The SD scenario closes that gap.

5.1.2 Datasets and Models

We evaluate our framework on four datasets spanning different resolutions and scales: CIFAR-10 and CIFAR-100 (both 32×32, 50k training images with 5k reserved as the watermark set), STL-10 (96×96, 5k training images with 1k as watermark), and CelebA (64×64, 162k training images with 5k as watermark). All splits are deterministic and stratified with a fixed seed to ensure reproducibility.

For the Stable Diffusion experiments, the owner's private training set consists of 1,000 images from COCO 2014; per-image captions are retained for use in latent-space scoring (Section 5.1.3). Adversaries fine-tune on a disjoint COCO subset (also 1,000 images) plus a synthetic-image set; details appear in Section 5.1.4.

5.1.3 Owner Training Protocols

For each dataset, we train a DDIM model using a U-Net backbone with 128 base channels, a cosine noise schedule, and the AdamW optimizer at a learning rate of 2×10^{-4} with EMA decay 0.9999. Training runs for 400k iterations in all cases. The Gaussian QR component uses an ensemble of $B{=}50$ models, each trained on a bootstrap sample of the auxiliary non-member data. Table 1 summarizes the key hyperparameters; full architectural and optimization details are deferred to Appendix A.

For Stable Diffusion v1.4, owner models are produced by fine-tuning the base checkpoint on the COCO subset described in Section 5.1.2. We consider three owner configurations: LoRA at rank $r{=}64$ (33.5M trainable parameters, 3.9% of the UNet), LoRA at rank $r{=}256$ (134M, 15.6%), and full UNet fine-tuning (860M, 100%). Common settings are AdamW, cosine schedule with 500 warmup steps, batch size 4, fp16 mixed precision, seed 42; per-variant differences (steps, learning rate, augmentation) appear in Table 10 of Appendix A.1. The t-error is scored in VAE latent space rather than pixel space; the latent-vs-pixel ablation that motivates this choice is in Section 5.6.

Table 1: Key hyperparameters for reproducibility. The DDIM and Gaussian QR rows are described in this subsection; the Stable Diffusion rows are split between owner training (described here) and adversary fine-tuning (described in Section 5.1.4 and Section 5.4).

Component	Parameter	Value
DDIM	Iterations	400k
	Batch size / LR	128 / 2×10^{-4}
	EMA decay	0.9999
Gaussian QR	Ensemble B	50
	Bootstrap ratio	0.8
	Epochs / patience	50 / 10
Stable Diffusion	Base model	SD v1.4
	LoRA rank / α	64 / 64, 256 / 256
	Fine-tune steps / LR	10,000 / 1×10^{-4}
	Private set size	1,000 COCO 2014
	Adversary fine-tune (steps / lr)	2,000 / 5×10^{-5}
	Adversary LoRA target / rank	cross-attention / 64 or 256
	Adversary loss	ϵ-prediction MSE
	Adversary training data	1,000 COCO images disjoint from $\mathcal{W}$

5.1.4 Adversary Protocols

We consider four post-theft attacks on the DDIM owner. **MMD fine-tuning** runs 500 iterations of CLIP-feature distribution matching (cubic polynomial kernel over CLIP ViT-B/32 embeddings, $\mathcal{D}_{\text{train}}$ as the target distribution, lr 5×10^{-6}, EMA 0.9999) and represents an adversary attempting to alter the model's generation distribution while leaving the architecture intact. **SGD fine-tuning** (500 iterations, lr 5×10^{-6}, batch 128) on clean data represents a domain-preserving adversarial fine-tune. **Gaussian weight noise** at $\sigma \in \{10^{-3}, 10^{-2}\}$ applied in-place to all parameter tensors models a quantization or noise-perturbation adversary. The four attacks span the most plausible post-theft modifications under our threat model; quantitative results are in Section 5.4.

For Stable Diffusion adversaries, a single 2,000-step LoRA fine-tune protocol (lr 5×10^{-5}, cross-attention modules only, ϵ-prediction MSE loss) is instantiated across two axes. The data axis yields two threat scenarios: **domain-shift** on a disjoint 1,000-image COCO subset (the adversary adapts the stolen model to a different but distributionally similar dataset), and **task-shift** on synthetic data (the adversary repurposes the model toward a generation task different from the owner's). The rank axis varies the adversary's LoRA rank, $r \in \{64, 256\}$. The resulting 2×2 grid of (rank, scenario) adversary configurations is applied to each of the three owner configurations from Section 5.1.3. Verification results across owners and scenarios are reported in Table 3 of Section 5.4 (rank dimension collapsed); the SleeperMark comparison under the same adversary protocol is in Section 5.5.

5.1.5 Baselines and Reference Models

The verification protocol uses two categories of external models: public reference checkpoints serving as the null hypothesis in the C2 separation criterion (used throughout Sections 5.2–5.4), and comparison methods used in the controlled baseline evaluation of Section 5.5.

Public reference baselines. As public baselines for the verification protocol, we use pre-trained checkpoints from HuggingFace: `google/ddpm-cifar10-32` serves as the reference for both CIFAR-10 and CIFAR-100, `google/ddpm-ema-bedroom-256` (resized) for STL-10, and `google/ddpm-celebahq-256` for CelebA. These baselines are trained on disjoint data from our watermark sets and represent the null hypothesis in the two-point verification protocol: a model that has never seen the owner's private evidence should exhibit substantially higher reconstruction error on that set.

Comparison methods. For the controlled baseline comparison in Section 5.5, we compare MiO against three diffusion-model IP protection methods: **WDM** [Peng et al., 2023], a training-time watermarking method that learns a secondary watermark diffusion process; **Zhao et al.** [Zhao et al., 2023], which embeds StegaStamp fingerprints into training images before EDM training; and

Table 2: Multi-reference ownership verification across 4 datasets. Each owner model is tested against 3 public reference models grouped into semantic-similar DDPM / cross-domain LSUN / random initialization. Verification requires *every* reference to satisfy $|d| > 2.0$ (ALL-must-pass). All 12 (dataset, reference) pairs satisfy the threshold; verification passes on all four datasets. The rightmost two columns report the minimum $|d|$ achieved per dataset and the multiplicative margin above the recommended threshold $|d| > 2.0$. All pairwise $p < 10^{-100}$. †Conservative reference (LSUN-Church, replacing LSUN-Bedroom to address domain-match concerns from prior review).

Dataset	Semantic-similar DDPM $\lvert d\rvert$	Cross-domain LSUN DDPM $\lvert d\rvert$	Random init $\lvert d\rvert$	Min $\lvert d\rvert$	Margin
CIFAR-10	20.7	31.6	45.1	20.7	10.4×
CIFAR-100	16.3	25.2	44.1	16.3	8.1×
STL-10	87.3	22.2†	41.5	22.2	11.1×
CelebA	20.5	23.5	63.0	20.5	10.2×

References per dataset: CIFAR-10/100 use `google/ddpm-cifar10-32`, `google/ddpm-bedroom-256`, random-32. STL-10 uses `google/ddpm-cifar10-32`, `google/ddpm-church-256`†, random-96. CelebA uses `google/ddpm-celebahq-256`, `google/ddpm-bedroom-256`, random-64.

CDI [Dubiński et al., 2024], an inference-time MIA-aggregation method. For Stable Diffusion, we additionally compare against **SleeperMark** [Wang et al., 2025], a trigger-prompt-coupled, LoRA-resistant watermark. Implementation details and hyperparameters appear in Appendix A.1.

5.1.6 Evaluation Metrics

We report three families of evaluation metrics.

Verification verdicts. The two-point criteria of Section 4.4 produce a binary $\{\text{VERIFIED}, \text{REJECTED}\}$ verdict per (owner, suspect) pair, together with the supporting C1 p-value and C2 effect size $|d|_{\text{C2}}$. A verdict of VERIFIED requires C1 ($p > 0.05$) and C2 ($|d|_{\text{C2}} > 2.0$ against every reference, with $p < 10^{-6}$).

Per-sample membership inference. To characterize the underlying signal, we report ROC-AUC and TPR at fixed FPR levels (0.1% and 0.01%) using the Gaussian QR ensemble margins of Section 4.3.

Effect sizes. Throughout, $|d|_{\text{C2}}$ denotes the paired Cohen's d as defined in Equation 15. For per-sample membership analyses (e.g., owner-level AUC), we additionally report unpaired Cohen's d between member and non-member score distributions, denoted $|d|_{\text{mem}}$.

5.2 RQ1: Verification Effectiveness

Can private member evidence sets distinguish owner models from public baselines?

Table 2 reports Cohen's d against three categories of public reference models (semantic-similar DDPM, cross-domain LSUN DDPM, and random initialization) under the ALL-must-pass criterion.

Results: Table 2 reports verification under the ALL-must-pass criterion: every public reference model must individually satisfy $|d| > 2.0$. We test each owner model against 3 reference models spanning a semantic-similar DDPM, a cross-domain LSUN DDPM, and a random initialization. The minimum $|d|$ across all references ranges from 16.3 (CIFAR-100, weakest dataset) to 22.2 (STL-10 with the LSUN-Church reference), exceeding the threshold by $8\times$ or more across all references. For STL-10, we use LSUN-Church as a conservative replacement for LSUN-Bedroom, which a prior reviewer flagged as a potential domain-match confound; the church reference yields $|d| = 22.2$, still $11\times$ above the $|d| > 2.0$ threshold. These margins confirm that the verification signal is an intrinsic property of the training data, not an artifact of a particular reference model choice. The two rightmost columns of Table 2 additionally report each dataset's minimum $|d|$ and its multiplicative margin above the recommended threshold $|d| > 2.0$; the smallest margin (CIFAR-100 at $8.1\times$) leaves substantial headroom, confirming that results are not artifacts of threshold tuning.

Table 3: Ownership verification on SD v1.4 (1,000 private COCO images). C1: consistency ($p > 0.05$). C2: separation ($|d| > 2.0$). Adversaries: domain-shift fine-tuning on disjoint COCO and task-shift on synthetic data. C2 ($|d|$): effect size $|d|_{\text{C2}} = |\text{mean}(\delta_{\text{C2}}(\mathcal{W}))/\text{std}(\delta_{\text{C2}}(\mathcal{W}))|$ as defined in Equation 15. $|d|_{\text{mem}}$: effect size between the owner's score distributions on member and non-member samples in $\mathcal{W}$, via standard absolute Cohen's d.

Owner	Adversary	C1	C2 ($\lvert d\rvert$)	AUC	$\lvert d\rvert_{\text{mem}}$	Verdict
LoRA $r=64$	Domain-shift	✓ (p=0.73)	✓ (2.43)	0.986	2.83	✓
LoRA $r=64$	Task-shift	✓ (p=0.34)	✓ (2.43)	0.980	2.70	✓
LoRA $r=256$	Domain-shift	✓ (p=0.54)	✓ (2.55)	0.998	3.51	✓
LoRA $r=256$	Task-shift	✓ (p=0.22)	✓ (2.55)	0.997	3.46	✓
Full FT	Domain-shift	✓ (p=0.71)	✓ (2.38)	0.997	3.22	✓
Full FT	Task-shift	✓ (p=0.18)	✓ (2.38)	0.995	3.13	✓

Consistency (C1). Owner and Model B produce statistically indistinguishable mean t-error on $\mathcal{W}$ across all four datasets: CIFAR-10: 28.73 vs. 28.74 ($p = 0.98$); CIFAR-100: 33.35 vs. 33.34 ($p = 0.98$); STL-10: 70.92 vs. 71.38 ($p = 0.71$); CelebA: 63.79 vs. 63.80 ($p = 0.97$). All p-values exceed the 0.05 acceptance threshold, confirming that Model B retains the memorization signature of its owner.

Stable Diffusion verification. We now apply the verification protocol to the SD scenario described in Section 5.1.1, where the owner is SD v1.4 with private fine-tuning and the reference is the original SD v1.4 base. Table 3 reports ownership verification across three owner configurations. The paired form of C2 in Section 4.4 (Equation 15) is essential here: owner and reference share the SD base, so their raw t-errors are nearly identical, but $\delta_{\text{C2}}(x)$ cancels the shared per-image variance and exposes the systematic memorization gap. C1 uses the same unpaired Welch test on raw owner-vs-suspect scores as in the DDIM setting (Algorithm 2).

For reference, owner-level membership detection on the watermark set $\mathcal{W}$ (1,000 members vs. 1,000 non-members, global evaluation aggregated across both adversary fine-tuning directions) achieves AUC $= 0.995/0.999/0.999$ and $|d|_{\text{mem}} = 3.26/3.93/3.41$ for LoRA r=64 / LoRA r=256 / Full FT respectively. Per-adversary verification results are aggregated using the paired-C2 statistic of Equation 15, which is invariant to adversary fine-tuning direction.

All three owner configurations satisfy both criteria across both adversary scenarios ($|d|_{\text{C2}} \in \{2.38, 2.43, 2.55\}$, $p_{\text{C1}} > 0.05$ uniformly), producing a ✓ verdict on all six (owner, adversary) cells of Table 3.

5.3 RQ2: False Positive Control

The C2 separation criterion requires significance at $p < 10^{-6}$ against every reference, and Section 5.2 demonstrated that owner models satisfy this threshold by large margins. The complementary question this section asks is whether the protocol *rejects* candidate models that are *not* the owner: given an arbitrary FPR budget α, can the protocol be calibrated so that an independent (non-derivative) model is mistaken for the owner with probability at most α? The answer rests on the analytical FPR control built into the Gaussian quantile regression of Section 4.3.

Recall that the Gaussian QR component models the conditional log-score distribution for non-members as a Gaussian with input-dependent mean and variance, parametrized by a network f_ψ trained on auxiliary non-member data (Section 4.3). Because the conditional distribution is Gaussian, the $(1-\alpha)$-quantile $\hat{q}_\tau(x) = \mu(x) + \sigma(x)\Phi^{-1}(\tau)$ with $\tau = 1 - \alpha$ admits a closed-form expression at any target FPR α without retraining. This is the analytical FPR control mechanism: the threshold separating "likely member" from "likely non-member" adapts to each input via $\mu(x), \sigma(x)$ rather than relying on a single global cutoff.

Table 4 reports per-sample membership inference performance under this analytical calibration. We measure ROC-AUC and TPR at fixed FPR levels of 0.1% and 0.01% on each dataset's owner model, with 95% bootstrap confidence intervals over 200 resamples. AUC remains above 0.96 on all four datasets, and at the most stringent FPR budget (0.01%, i.e., one false positive per ten thousand non-members), TPR remains above 0.55 uniformly. Analytical FPR control thus delivers practically

Table 4: Membership inference performance using Gaussian QR margins on DDIM owner models. TPR computed at fixed FPR with 95% bootstrap CI (200 iterations).

Dataset	AUC	TPR@0.1%	TPR@0.01%
CIFAR-10	0.987 ± 0.002	0.89 ± 0.03	0.71 ± 0.05
CIFAR-100	0.982 ± 0.003	0.85 ± 0.04	0.65 ± 0.06
STL-10	0.968 ± 0.005	0.78 ± 0.05	0.55 ± 0.07
CelebA	0.975 ± 0.004	0.82 ± 0.04	0.60 ± 0.06

usable detection at the budgets relevant to ownership disputes (where false-positive rates of even 1% would be intolerable).

Scope of this analysis. The FPR control reported here is *analytical*: it derives from the Gaussian assumption in the QR model, and the empirical evidence in Table 4 validates the assumption only indirectly via the AUC and TPR statistics. A direct empirical FPR audit—running a battery of independently trained (non-derivative) models through the verification protocol as candidate owners and measuring the per-trial rejection rate—would constitute a stronger guarantee. We leave that audit to future work; the current paper relies on the analytical control plus the AUC/TPR validation.

5.4 RQ3: Post-Theft Robustness

A practical adversary who obtains a stolen model will typically fine-tune it to a new domain or task before deployment. The ownership signal must survive such adaptation. To evaluate this, we simulate post-theft modification via MMD fine-tuning [Gretton et al., 2012]. MMD fine-tuning contrasts CLIP features of the model's DDIM-generated samples against those of real training images sampled fresh each step from $\mathcal{D}_{\text{train}}$ (which includes $\mathcal{W}$), via a cubic polynomial kernel. Starting from Model A, we fine-tune for 500 iterations using 10-step DDIM sampling, CLIP ViT-B/32 as the feature encoder, a learning rate of 5×10^{-6}, and EMA decay 0.9999.

The resulting Model B retains nearly identical t-error to Model A on the watermark set (Table 2), and both verification criteria remain satisfied across every dataset. This robustness is not coincidental: because the loss operates at the distribution level rather than imposing a per-image reconstruction objective on $\mathcal{W}$, the memorization gap on individual $\mathcal{W}$ examples persists. While WDM and Zhao also withstand these attacks in our evaluation (Table 5), MiO achieves this robustness without the training-time cost and FID degradation that those methods incur (Table 6). We note, however, that an adversary who retrains from scratch or specifically targets memorization erasure falls outside our threat model.

Table 5 stress-tests the three diffusion-model baselines under four post-theft attacks: MMD fine-tuning and standard SGD fine-tuning (both 500 iterations at $\text{lr}=5\times10^{-6}$ on clean CIFAR-10), and additive Gaussian weight noise at $\sigma=10^{-3}$ and $\sigma=10^{-2}$. All three methods pass every measured attack cell at our chosen strengths; MiO's distinguishing value therefore lies not in exclusive robustness but in delivering comparable robustness with no training overhead, no FID cost, and no modification to the target model's weights (Table 6). CDI's robustness under fine-tuning is not reported in the original paper and was not re-run on adversary checkpoints here; we therefore omit CDI from Table 5 and defer that evaluation to future work.

Stable Diffusion adversary robustness. We evaluate adversary robustness in the SD scenario by applying the 2×2 (rank, scenario) grid of adversaries from Section 5.1.4 to each owner configuration. The full verification verdict matrix is the SD column of Table 3 reported in Section 5.2: all three owner configurations (LoRA r=64, LoRA r=256, Full FT) retain a ✓ verdict across both adversary scenarios, with C2 effect sizes $|d|_{\text{C2}} \in \{2.38, 2.43, 2.55\}$ depending only on the owner configuration (the paired-C2 statistic is invariant to adversary fine-tuning direction by construction). The SD scenario therefore exhibits the same robustness pattern as the DDIM-from-scratch setting: post-theft adaptation does not displace the memorization signal in the watermark set.

Table 5: Robustness of watermark and ownership verification under four post-theft attacks on CIFAR-10. ✓ indicates the method passes its native pass/fail threshold; "—" indicates not applicable. Pass thresholds: pixel accuracy > 0.9 for WDM (on the WDP-extracted watermark image); bit accuracy > 0.9 for Zhao (on $N = 1000$ Heun-18 generated samples decoded with the ground-truth 64-bit StegaStamp fingerprint); two-point verification (C1: $p > 0.05$, C2: $|d| > 2.0$) for MiO. WDM/Zhao thresholds are MiO-defined; the original papers do not specify robustness thresholds. Attack configurations: MMD-FT and SGD-FT each run for 500 iterations at $\text{lr} = 5 \times 10^{-6}$ on clean CIFAR-10. SGD-FT uses each method's native training objective (ε-prediction MSE for the DDPM-parameterized WDM and MiO; σ-weighted $\hat{x}_0$-prediction MSE for Zhao's EDM backbone) to avoid introducing a parameterization mismatch. Gaussian weight noise is applied in-place to all parameter tensors of each method's native network. ‡Zhao × MMD-FT requires a Heun-based differentiable sampler not implemented in our DDIM-based MMD fine-tuning loop; deferred to future work.

Method	MMD-FT	SGD-FT	$\sigma = 10^{-3}$	$\sigma = 10^{-2}$
WDM	✓	✓	✓	✓
Zhao†	—‡	✓	✓	✓
MiO	✓	✓	✓	✓

†Uses EDM backbone (not DDPM/DDIM).

5.5 RQ4: Controlled Baseline Comparison

Whereas Section 5.4 shows that MiO matches WDM and Zhao under post-theft attacks, this section focuses on the orthogonal question of capability cost: training overhead, FID degradation, and weight modification. We compare MiO against three existing methods on CIFAR-10: two training-time watermarking methods (WDM [Peng et al., 2023], which learns a secondary watermark diffusion process alongside the standard DDPM objective, and Zhao et al. [Zhao et al., 2023], which embeds a StegaStamp bit-string into training images before fitting an EDM model) and one inference-time MIA-aggregation method (CDI [Dubiński et al., 2024]), which aggregates seven complementary membership inference attacks (denoising loss, SecMI-stat, PIA, PIAN, gradient masking, noise optimization, multiple-loss) through a logistic-regression meta-classifier. All three baselines have publicly available official implementations. WDM and Zhao are retrained on the same 50K CIFAR-10 images so that any differences reflect the methods themselves rather than data or scale effects; CDI is evaluated on the public `google/ddpm-cifar10-32` checkpoint using the authors' released codebase [Dubiński et al., 2024] with three trivial patches required for the aggregation stage to execute end-to-end (Appendix A.1); 5K members + 5K non-members under the default 7-attack configuration. We exclude DeepMarks [Chen et al., 2019], whose GMM-regularised weight fingerprinting was designed for discriminative classifiers and has no reference implementation for generative architectures.

Table 6 summarises the four methods along four capability dimensions: whether the method modifies the target model's weights, the additional training cost it incurs, its FID cost (the generation-quality penalty attributable to the method), and its per-sample membership AUC. WDM and Zhao both require a full (re)training pass with a modified objective and introduce a measurable FID cost (+7.21 and +3.07 respectively). CDI and MiO are both inference-only: they leave the target model's weights unchanged, so their FID cost is zero by construction and they incur no training overhead.

On the one dimension where CDI and MiO are directly comparable— per-sample membership AUC on CIFAR-10—MiO achieves 0.987 against CDI's 0.586, a substantial gap even after accounting for the fact that CDI and MiO evaluate on different base checkpoints (`google/ddpm-cifar10-32` for CDI; the owner's custom DDIM for MiO). The gap reflects the structural advantage of operating against a private member evidence set held by the owner: when the defender controls a known member subset, per-sample detection becomes tractable at stringent false-positive rates, whereas open-ended dataset inference operates in a harder setting.

SleeperMark comparison on Stable Diffusion. Stable Diffusion has its own dedicated watermarking baseline, SleeperMark [Wang et al., 2025], designed for LoRA-adapted SD. We compare under the 2,000-step LoRA adversary protocol described in Section 5.1.4: SleeperMark passes if the recovered bit-accuracy exceeds 0.9, and MiO passes if both two-point criteria (C1 + C2) hold. Both methods pass their respective thresholds for LoRA r=64 and r=256 adversaries (Table 7);

Table 6: Capability matrix of diffusion-model IP protection methods on CIFAR-10. ✓/✗ indicate whether the method has the property; — indicates the column is not applicable to the method. FID cost values for WDM and Zhao are reported under their original sampling protocols (WDM: 50-step DDIM; Zhao: 18-step Heun on EDM backbone). CDI and MiO are inference-only and do not alter model weights; their FID cost is 0 by construction. Per-sample AUC is reported on CIFAR-10: CDI evaluated on `google/ddpm-cifar10-32` (5K members + 5K non-members); MiO evaluated on its owner DDIM with a 5K private watermark set.

Method	Modifies weights?	Training overhead	FID cost	Per-sample AUC
WDM	✓	$\sim 1\times$	$+7.21$	—
Zhao[†]	✓	$\sim 1\times$	$+3.07$	—
CDI	✗	0	0	0.586
MiO (Ours)	✗	**0**	**0**	**0.987**

[†]Uses EDM backbone (not DDPM/DDIM).

Table 7: Watermark robustness on SD v1.4: SleeperMark vs. MiO. **Threat**: adversary obtains the watermarked base model and applies 2,000-step LoRA fine-tuning (lr 5×10^{-5}, cross-attention modules, ϵ-prediction MSE on 1,000 disjoint COCO images). **Success criterion**: SleeperMark passes if bit-accuracy > 0.9; MiO passes if two-point criteria (C1 + C2) are satisfied. Both methods pass their respective thresholds under LoRA adaptation.

Method	LoRA $r=64$	LoRA $r=256$
SleeperMark	✓	✓
MiO (Ours)	✓	✓

MiO matches SleeperMark's verdict without modifying the owner's model weights or incurring training-time watermarking cost.

5.6 Ablations and Robustness Studies

Several design choices underlie the verification pipeline—how to collapse a sequence of per-timestep errors into a single score, how many timesteps are needed for that score to be reliable, whether the Gaussian assumption in our quantile regression is worth the modelling commitment, and (for the SD scenario) the choice of measurement space and prompt conditioning. We address each in turn on CIFAR-10 (and on SD v1.4 for the latent-vs-pixel choice); ensemble size and wall-clock breakdowns appear in Appendix B.

Table 8 compares four ways of summarising the t-error sequence. The mean, which weights every timestep equally, washes out the signal at noise levels where the memorisation gap is widest; the median is slightly more selective but still averages over a broad middle band. Moving to lower percentiles helps because the ownership signal concentrates at moderate noise levels—timesteps at which the model reconstructs its own training data with noticeably lower error than unseen samples. Q25 captures this sweet spot, reaching $|d| = 23.93$. Pushing further to Q10 begins to pick up outlier timesteps whose high variance hurts more than their selectivity helps, and separation drops back slightly.

The number of sampled timesteps K matters less than one might expect. Varying K from 10 to 100, we find that $|d|$ plateaus once $K \geq 25$ and changes negligibly beyond $K = 50$. We therefore fix $K = 50$ throughout; practitioners under tighter compute budgets can halve this to $K = 25$ with only marginal loss.

Finally, we ask whether the Gaussian distributional assumption in our quantile regression buys anything over a distribution-free alternative. Pinball-loss QR optimises directly for a target quantile τ without assuming a parametric form, but it requires a separate model for every τ of interest. On CIFAR-10, at $\tau \in \{0.999, 0.9999\}$ (FPR $= 0.1\%$ and 0.01%), the two approaches yield comparable TPR. The practical difference is flexibility: Gaussian QR trains once and evaluates any τ in closed form, cutting both the parameter count and the retraining burden roughly in half. For a verifier who

Table 8: Aggregation ablation on CIFAR-10. Lower percentiles sharpen separation up to a point; Q25 offers the best trade-off before outlier sensitivity sets in. Ablation conducted on CIFAR-10; SD results use Q25 and are consistent with this ranking (see Section 5.5).

Aggregation	$\lvert d \rvert$
Mean	18.2
Q10	21.5
Q25 (ours)	**23.93**
Median	19.8

may need to assess ownership claims at several significance levels—exactly the setting envisioned by our protocol—this is a meaningful convenience.

Stable Diffusion: latent vs pixel space. SD operates in VAE latent space, so the t-error can be measured either in latent space (before the decoder) or in pixel space (after decoding); prompt conditioning likewise admits two choices (per-image caption or empty). On the SD v1.4 owner model, latent-space scoring with per-image caption conditioning yields AUC$=0.9956$, compared to 0.959 for pixel-space scoring—a 3.7 percentage-point gap. Two factors explain the difference. First, the VAE decode step introduces a lossy reconstruction floor that dilutes the membership signal in pixel space. Second, SD learned image-caption associations during training; using empty prompts bypasses the cross-attention pathway where LoRA adapters operate, reducing the observable memorization gap. All SD experiments in this paper use the latent-plus-caption configuration.

6 Discussion and Limitations

We frame scope conditions rather than limitations.

White-box Access: Our method applies where model weights are available, standard in IP litigation (courts compel disclosure), regulatory audits (GDPR/AI Act), partnership disputes, and increasingly common open-weight models (LLaMA, Mistral).

Watermark Retention: Verification requires the owner to retain the watermark set, analogous to keeping receipts for warranty claims.

Retroactive Claims: Our protocol assumes the owner's evidence set $\mathcal{W}$ is established through normal training-data curation prior to any dispute, not selected post-hoc from a released training corpus. The combinatorial argument in Section 3 bounds the probability that an adversary observing $\mathcal{D}_{\text{train}}$ ex post can guess the exact $|\mathcal{W}|$-subset (e.g., 10^{-5000} for the CIFAR-10 setting), but this bound is meaningful only when $\mathcal{W}$ predates the dispute and the owner can demonstrate its prior existence through normal-course evidence (training logs, dataset versioning, internal documentation). The protocol does not by itself adjudicate the temporal authenticity of $\mathcal{W}$; that determination is delegated to the same evidentiary processes that govern other intellectual-property disputes.

Baseline Availability: Results are strongest with domain-matched public baselines; for novel domains, general-purpose baselines provide conservative bounds.

Gaussian Assumption: The Gaussian model provides a principled approximation; empirical calibration shows quantile estimates remain accurate under moderate misspecification.

Distillation Boundary: Our threat model assumes the adversary retains the original training distribution via weight-preserving operations (fine-tuning, pruning, quantisation). Knowledge distillation—generating synthetic data from the stolen model and training a fresh student—violates this assumption and verification correctly reports no ownership (FAIL via C1 consistency rejection; Appendix E). This is a principled boundary shared by proactive watermarking methods: WDM and Zhao embed signals into model weights, which distillation with fresh initialisation erases equally.

What This Work Does NOT Address: API-only verification (we focus on white-box), full retraining adversaries (our threat model assumes fine-tuning), and adversaries specifically optimizing to remove memorization signals.

7 Conclusion

We presented a novel ownership verification framework for diffusion models using t-error based reconstruction disparity and Gaussian Quantile Regression. By formalizing the distinction between membership inference and ownership verification, we established that public baselines and statistical hypothesis testing are *necessary* components for ownership claims. Our two-point verification criteria (consistency, separation) provide rigorous, reproducible verification. Experiments across four datasets demonstrate $|d| > 16$ separation against every reference model, with criteria satisfied even after adversarial MMD fine-tuning.

References

Yossi Adi, Carsten Baum, Moustapha Cissé, Benny Pinkas, and Joseph Keshet. Turning your weakness into a strength: Watermarking deep neural networks by backdooring. In *USENIX Security Symposium*, pages 1615–1631, 2018.

Martin Bertran, Shuai Tang, Michael Kearns, Jamie Morgenstern, Aaron Roth, and Zhiwei Steven Wu. Scalable membership inference attacks via quantile regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Nicholas Carlini, Jamie Hayes, Milad Nasr, Matthew Jagielski, Vikash Sehwag, Florian Tramèr, Borja Balle, Daphne Ippolito, and Eric Wallace. Extracting training data from diffusion models. In *Proceedings of the 32nd USENIX Security Symposium (USENIX Security 2023)*, pages 5253–5270, 2023.

Huili Chen, Bita Darvish Rohani, and Farinaz Koushanfar. DeepMarks: A secure fingerprinting framework for digital rights management of deep learning models. In *ACM International Conference on Multimedia Retrieval (ICMR)*, pages 105–113, 2019.

Hai Ci, Pei Yang, Yiren Song, and Mike Zheng Shou. Ringid: Rethinking tree-ring watermarking for enhanced multi-key identification. In *Computer Vision – ECCV 2024*, volume 15086 of *Lecture Notes in Computer Science*, pages 338–354, 2024.

Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 2nd edition, 1988.

Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 8780–8794, 2021.

Jiaxin Duan, Fanbo Kong, Ruoxi Wang, Karan Gu, et al. Diffusion model membership inference via reverse process discrepancy. In *International Conference on Machine Learning (ICML)*, 2023.

Jan Dubiński, Antoni Kowalczuk, Franziska Boenisch, and Adam Dziedzic. CDI: Copyrighted data identification in diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

Pierre Fernandez, Guillaume Couairon, Hervé Jégou, Matthijs Douze, and Teddy Furon. The stable signature: Rooting watermarks in latent diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Arthur Gretton, Karsten M Borgwardt, Malte J Rasch, Bernhard Schölkopf, Alexander Smola, et al. A kernel two-sample test. *Journal of Machine Learning Research (JMLR)*, 13(1):723–773, 2012.

Hailong Hu and Jun Pang. Loss and likelihood based membership inference of diffusion models. In *Proceedings of the 26th Information Security Conference (ISC)*, volume 14411 of *Lecture Notes in Computer Science*, pages 121–141. Springer, 2023.

Changhoon Kim, Kyle Min, Maitreya Patel, Sheng Cheng, and Yezhou Yang. WOUAF: Weight modulation for user attribution and fingerprinting in text-to-image diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8974–8983, 2024.

Fei Kong, Jinhao Duan, Ruipeng Ma, Heng Tao Shen, Xiaoshuang Shi, Xiaofeng Zhu, and Kaidi Xu. An efficient membership inference attack for the diffusion model by proximal initialization. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024.

Kecen Li, Zhicong Huang, Xinwen Hou, and Cheng Hong. Gaussmarker: Robust dual-domain watermark for diffusion models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.

Tomoya Matsumoto, Takayuki Miura, and Naoto Yanai. Membership inference attacks against diffusion models. In *IEEE Security and Privacy Workshops (SPW)*, pages 77–83, 2023.

Sen Peng, Yufei Nie, Yongzhen Zhang, and Jie Xiong. Protecting the intellectual property of diffusion models by the watermark diffusion process. *arXiv preprint arXiv:2306.03436*, 2023.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2022.

Ahmed Salem, Yang Zhang, Mathias Humbert, Pascal Berrang, Mario Fritz, and Michael Backes. ML-Leaks: Model and data independent membership inference attacks and defenses on machine learning models. In *Network and Distributed System Security Symposium (NDSS)*, 2019.

Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *IEEE Symposium on Security and Privacy (S&P)*, pages 3–18, 2017.

Gowthami Somepalli, Vasu Singla, Micah Goldblum, Jonas Geiping, and Tom Goldstein. Diffusion art or digital forgery? investigating data replication in diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6048–6058, 2023.

Shuai Tang, Zhiwei Steven Wu, Sergül Aydöre, Michael Kearns, and Aaron Roth. Membership inference attacks on diffusion models via quantile regression. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, 2024.

Zilan Wang, Junfeng Guo, Jiacheng Zhu, Yiming Li, Heng Huang, and Muhao Chen. Sleepermark: Towards robust watermark against fine-tuning text-to-image diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

Yuxin Wen, John Kirchenbauer, Jonas Geiping, and Tom Goldstein. Tree-ring watermarks: Fingerprints for diffusion images that are invisible and robust. *arXiv preprint arXiv:2305.20030*, 2023.

Samuel Yeom, Irene Giacomelli, Matt Fredrikson, and Somesh Jha. Privacy risk in machine learning: Analyzing the connection to overfitting. In *IEEE Computer Security Foundations Symposium (CSF)*, pages 268–282, 2018.

Jialong Zhang, Zhongshu Gu, Jiyong Jang, Hui Wu, Marc Ph Stoecklin, Heqing Huang, and Ian Molloy. Protecting intellectual property of deep neural networks with watermarking. In *ACM Asia Conference on Computer and Communications Security (ASIACCS)*, pages 159–172, 2018.

Yunqing Zhao, Tianyu Pang, Chao Du, Xiao Yang, Ngai-Man Cheung, and Min Lin. A recipe for watermarking diffusion models. *arXiv preprint arXiv:2303.10137*, 2023.

A Additional Experimental Details

A.1 Hyperparameters

A.1.1 Owner Model Training

DDIM owner training. We train DDIM owner models for the four datasets in Section 5.1 using the configuration in Table 9.

Table 9: Training hyperparameters for DDIM models.

Parameter	Value
Total iterations	400,000
Batch size	128
Learning rate	2×10^{-4}
Optimizer	AdamW
Weight decay	0.0
EMA decay	0.9999
Noise schedule	Cosine
Timesteps T	1000
U-Net Architecture	
Base channels	128
Channel multipliers	[1, 2, 2, 2]
Attention resolutions	[16]
Dropout	0.1

Stable Diffusion fine-tuning. All three configurations share the same 1,000-image private subset of COCO 2014 as the watermark set $\mathcal{W}$; no per-configuration data resampling. We fine-tune SD v1.4 from the official base checkpoint. Per-variant differences are summarized in Table 10.

Table 10: Stable Diffusion owner fine-tuning configurations. Per-variant differences are listed above the rule; common settings shared by all three variants are listed below.

Parameter	LoRA r=64 (A6)	LoRA r=256 (A8)	Full FT (A7)
Trainable params	33.5M (3.9%)	134M (15.6%)	860M (100%)
Steps	20,000	20,000	10,000
Learning rate	1×10^{-4}	1×10^{-4}	5×10^{-6}
LoRA rank / α	64 / 64	256 / 256	—
LoRA target modules	cross-attn	cross-attn	—
Random horizontal flip	yes	no	no
Common settings (all three variants)			
Base model: SD v1.4	Optimizer: AdamW	Batch size: 4	Gradient accumulation: 1
Resolution: 512×512 resize with center crop		Captions: per-image COCO	
LR schedule: cosine with 500 warmup steps		Mixed precision: fp16	Seed: 42

A.1.2 Gaussian QR Training

Table 11: Gaussian QR training hyperparameters.

Parameter	Value
Backbone	ResNet-18 (pretrained)
Ensemble size B	50
Bootstrap ratio	0.8
Epochs per model	50
Batch size	256
Learning rate	0.001
LR schedule	Cosine annealing
Early stopping patience	10 epochs

A.1.3 Adversary Fine-tuning

The adversary applies LoRA fine-tuning to either the owner's fine-tuned SD model (Table 3) or the SleeperMark watermarked checkpoint (Table 7). Identical settings in both cases: 2,000 steps, AdamW (weight decay 10^{-2}), lr 5×10^{-5}, batch size 4, gradient clipping at norm 1.0, ϵ-prediction

MSE loss on 1,000 disjoint COCO images, seed 42. LoRA targets cross-attention modules {`to_q`, `to_k`, `to_v`, `to_out.0`} with rank/$\alpha \in \{64/64,\ 256/256\}$.

A.1.4 Generative Baseline Training: WDM and Zhao

Table 12 summarizes the training configurations for the two generative-model baselines (WDM and Zhao). Method-specific watermark mechanisms are described after the table.

Table 12: Training configurations for generative baselines (CIFAR-10).

Parameter	WDM	Zhao (EDM stage)
Architecture	U-Net 128ch	SongUNet (ddpmpp) 128ch
Channel multipliers	$[1, 2, 2, 2]$	$[2, 2, 2]$
Residual blocks/stage	3	4
Attention resolutions	$\{16, 8\}$, 4 heads	$\{16\}$
Activation / Norm	SiLU / GroupNorm(32)	(default EDM)
Dropout	0.1	0.13
Noise schedule	linear β, T=1000	EDMPrecond
Prediction	ϵ	$\hat{x}_0$ (σ-weighted MSE)
Training duration	300,000 steps	15 Mimg (early-stopped)[†]
Batch size	32	512 (8 GPUs $\times$ 64)
Learning rate	1×10^{-4}	1×10^{-3}
EMA decay	0.999	0.5
Optimizer	AdamW	AdamW
Augmentation prob.	—	0.12
Seed	42	42
FID sampling	DDIM 50 steps	Heun 18 steps

[†]Original Zhao paper trains to 200 Mimg; full duration would require approximately 57 single-GPU-days. The 15 Mimg checkpoint achieves 100% fingerprint bit accuracy, sufficient for the baseline comparison.

WDM watermark configuration. Watermark blend factor $\gamma_1 = 0.8$, training loss weight $\gamma_2 = 0.1$ on watermarked samples ($\gamma_2 = 0.2$ at extraction), watermark seed 1998, fixed trigger image. Loss: $\mathcal{L} = \mathrm{MSE}(\epsilon) + \gamma_2 \cdot \mathrm{MSE}(\epsilon_{\text{wm}})$.

Zhao three-stage pipeline. **Stage 1**: train StegaStamp encoder/decoder at 32×32 resolution, batch 64, 100 epochs, 64-bit fingerprint. **Stage 2**: embed a fixed 64-bit fingerprint (seed 42) into all 50K CIFAR-10 training images. **Stage 3**: train EDM with the configuration in Table 12.

A.1.5 SleeperMark Baseline

We use the pretrained SleeperMark U-Net checkpoint released by the authors (approximately 3.3 GB). No SleeperMark training is performed in our pipeline; the Table 7 results measure robustness of the released checkpoint under the adversary protocol described above.

A.1.6 CDI Reproduction

We re-evaluated CDI [Dubiński et al., 2024] on `google/ddpm-cifar10-32` using the authors' codebase[1] on a single GPU (Python 3.8.18, PyTorch 2.4.1+cu121, diffusers 0.36.0; total wall-clock $\approx$49 minutes). **Configuration**: 5,000 members and 5,000 non-members sampled from CIFAR-10; batch size 100; seed 0; three-stage Hydra pipeline (features_extraction $\rightarrow$ scores_computation $\rightarrow$ evaluation). **Aggregation**: logistic regression (max_iter 1000) over the union of four score-based attacks (denoising loss, SecMI-stat, PIA, PIAN) and three feature-based attacks (gradient masking, noise optimization, multiple-loss). Individual attack AUCs ranged from 0.491 (PIAN) to 0.557 (noise optimization); the aggregated CDI AUC reported in Table 6 is 0.586.

Patches. Three trivial patches were required for the upstream `scores_computation/cdi.py` aggregation stage to execute end-to-end: a missing `import numpy as np`; a `train_dataset["data"]`

[1]`https://github.com/sprintml/copyrighted_data_identification`

→ `train_dataset.data` fix for the `MIDataset` class API; and a 3D→2D reshape for multi-measurement features. None affects the aggregation algorithm itself.

A.1.7 C2 Computation Details

The Stable Diffusion verification numbers in Table 3 are computed via Equation 15: for each adversary configuration, $\delta_{\text{C2}}(x) = s_{\text{tgt}}(x) - s_{\text{ref}}(x)$ is computed on $\mathcal{W}$ from the per-image score CSVs (one row per image, columns `score_tgt` and `score_ref`), and $|d|_{\text{C2}} = |\text{mean}(\delta_{\text{C2}})/\text{std}(\delta_{\text{C2}})|$ is computed directly. A standalone runner that reproduces the Table 3 C2 values from the released CSVs will be released with the camera-ready code.

A.2 Post-theft Attack Configurations

MMD fine-tuning. 500 iterations on Model A using a cubic polynomial kernel over CLIP ViT-B/32 (frozen) features. Optimizer: AdamW at lr 5×10^{-6}, batch size 128, EMA decay 0.9999. Generation pipeline: 10-step DDIM sampling from t_{start}=900. The MMD target distribution is $\mathcal{D}_{\text{train}}$ (which includes $\mathcal{W}$); see Section 5.4 for the rationale.

SGD fine-tuning. 500 iterations, lr 5×10^{-6}, batch 128, applied on clean CIFAR-10 using each method's native training objective: ϵ-prediction MSE for the DDPM-parameterized WDM and MiO; σ-weighted $\hat{x}_0$-prediction MSE for Zhao's EDM backbone. The choice of native objectives avoids introducing a parameterization mismatch.

Gaussian weight noise. Additive isotropic Gaussian noise of standard deviation $\sigma \in \{10^{-3}, 10^{-2}\}$ applied in-place to all parameter tensors of each method's native network. No retraining; the perturbed checkpoint is evaluated directly.

B Gaussian QR Ensemble Size Ablation

We vary the bagging ensemble size B (introduced in Section 4.3) from 1 to 100 and measure the standard deviation of the predicted quantile $\hat{q}_\tau(x)$ across five independent runs on the CIFAR-10 watermark set at $\tau = 0.999$ (FPR $= 0.1\%$). Table 13 reports the results. Prediction variance drops sharply from $B = 1$ to $B = 20$, continues to decrease up to $B = 50$, and plateaus beyond that point. We adopt $B = 50$ as the default, which offers a favorable trade-off between stability and computational cost. Since ensemble members are trained independently, scaling from $B = 50$ to any larger value is embarrassingly parallel and requires no inter-model communication.

Table 13: Effect of ensemble size B on quantile prediction stability (CIFAR-10, $\tau = 0.999$). Std. dev. computed over 5 independent runs.

Ensemble B	Std($\hat{q}_\tau$)	TPR @ FPR 0.1%
1	0.142	0.83 ± 0.05
10	0.068	0.87 ± 0.03
20	0.041	0.88 ± 0.03
50 (ours)	0.019	0.89 ± 0.03
100	0.016	0.89 ± 0.02

C Computational Resources

Table 14 summarizes the computational requirements of each component. The dominant cost is ensemble training (approximately 100 A100-hours for $B = 50$), which parallelizes trivially. At inference, t-error computation for 5,000 watermark samples takes roughly 10 minutes on a single A100, and ensemble quantile prediction adds less than 1 minute. In total, a single ownership query (from raw images to verdict) completes in under 15 minutes, making the protocol practical for on-demand verification.

By comparison, explicit watermarking methods such as WDM [Peng et al., 2023] require re-training the diffusion model with the watermark objective, which typically takes as long as the original model

training (on the order of days for the architectures we consider). MiO's Gaussian QR training is a one-time offline cost, and the owner model itself is never modified.

Table 14: Computational cost breakdown on CIFAR-10 (single NVIDIA A100).

Component	Time	Parameters
DDIM training (owner model)	$\sim$48 h	35.7M
Gaussian QR ($\times$1 model)	$\sim$2 h	11.2M
Gaussian QR ensemble (B=50)	$\sim$100 h†	560M total
Inference (per verification query)		
t-error scoring (5k samples)	$\sim$10 min	—
Ensemble prediction	$<$1 min	—
Two-point criteria	$<$1 s	—
Total query time	**$\sim$11 min**	—

† Embarrassingly parallel; wall-clock $\sim$2 h with 50 GPUs.

D Gaussian QR Training Details

Algorithm 3 Gaussian Quantile Regression Training

Require: Auxiliary non-member data $\mathcal{D}_{\text{aux}}$, ensemble size B, bootstrap ratio r
Ensure: Trained ensemble $\{f_\theta^{(b)}\}_{b=1}^B$
  for $b = 1$ to B **do**
    $\mathcal{D}_b \leftarrow$ BootstrapSample($\mathcal{D}_{\text{aux}}, r$) {80% with replacement}
    $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}} \leftarrow$ Split($\mathcal{D}_b, 0.9$)
    Initialize $f_\theta^{(b)}$ with ResNet-18 backbone
    repeat
      for batch $(x, s(x), \phi(x))$ in $\mathcal{D}_{\text{train}}$ **do**
        $y \leftarrow \log(1 + s(x))$
        $(\mu, \log\sigma) \leftarrow f_\theta^{(b)}(x, \phi(x))$
        $\mathcal{L} \leftarrow (y - \mu)^2/(2\sigma^2) + \log\sigma$
        Update θ via gradient descent
      end for
    until early stopping (patience = 10)
  end for
  return $\{f_\theta^{(b)}\}_{b=1}^B$

E Threat Model Boundary: Distillation

An adversary who generates a large synthetic dataset from the stolen model and trains a new model from scratch effectively performs *knowledge distillation*. Because the student is initialised independently and never sees the owner's training data directly, per-example memorisation of $\mathcal{W}$ is not expected to transfer.

Experiment. We generate 50,000 synthetic CIFAR-10 images from Model A via 10-step DDIM sampling and train a student model from scratch (100,000 iterations, batch size 128, fresh random initialisation). Table 15 reports t-error statistics on $\mathcal{W}$.

The student's mean t-error on $\mathcal{W}$ is 106.9, compared to 28.7 for the owner—a $3.72\times$ gap. The verification protocol correctly returns FAIL: the consistency criterion (C1) rejects because the student's score distribution differs significantly from the owner's ($p \approx 0$), indicating the student does not reproduce the owner's per-example memorisation pattern.

This outcome is expected and represents a principled boundary of the method rather than a failure mode. Distillation with fresh initialisation approximates full retraining: generating a sufficiently

Table 15: Distillation boundary experiment on CIFAR-10. The student model, trained on synthetic data from the owner, exhibits 3.72× higher t-error than the owner and fails the consistency criterion.

Model	Mean t-error	Std	Ratio vs. Owner
Model A (owner)	28.71	10.85	1.0×
Student (distilled)	106.90	36.14	3.72×
ddpm-cifar10 (reference)	703.84	38.50	24.5×
Random UNet (lower bound)	3037.31	65.75	105.8×

large synthetic dataset from a diffusion model and training a new model from scratch approaches the computational cost of the original training procedure, undermining the economic motivation for model theft. Critically, the protocol does not produce a false positive—it correctly reports that ownership cannot be established, which is the far more important guarantee.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction (Section 1) claim Cohen's $d > 18$ and error ratios $> 19\times$ across four datasets, with robustness under MMD fine-tuning. These are directly supported by Table 2 and Section 5.4. Limitations regarding white-box access and full retraining are stated in Section 6.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 explicitly discusses five scope conditions: white-box access requirement, watermark retention, baseline availability, the Gaussian assumption, and what the work does not address (API-only verification, full retraining adversaries, memorization erasure).

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The Gaussian QR formulation (Section 4.3) states its distributional assumption explicitly ($y \sim \mathcal{N}(\mu(x), \sigma^2(x))$) and derives the closed-form quantile threshold. The combinatorial bound on watermark guessing (Section 3) is a counting argument with all assumptions stated.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Table 1 and Appendix A provide all hyperparameters for DDIM training, Gaussian QR, and MMD fine-tuning. Dataset splits are deterministic with fixed seeds. Public baseline checkpoints are identified by their HuggingFace model IDs. Algorithm 1 and Algorithm 2 formalize the procedures.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: An anonymized code repository containing the MiO verification protocol, training scripts, and evaluation pipeline will be provided as supplementary material with this submission. See Section 5.1 for setup instructions and Appendix A for hyperparameters.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 5.1 specifies datasets, split sizes, seeds, and baseline models. Table 1 summarizes key hyperparameters. Appendix A provides full architectural details (U-Net channels, attention resolutions, dropout), optimizer settings (AdamW, learning rate, weight decay), and MMD fine-tuning configuration.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 4 reports 95% bootstrap confidence intervals (200 iterations) for TPR at fixed FPR. The verification protocol itself is grounded in hypothesis testing (p-values, Cohen's d), and all p-values for baseline separation are $< 10^{-100}$ (Table 2).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix C (Table 14) provides a detailed computational cost breakdown on NVIDIA A100 GPUs, including DDIM training time ($\sim$48h), Gaussian QR ensemble cost ($\sim$100 A100-hours, parallelizable to $\sim$2h wall-clock), and per-query inference time ($\sim$11 min).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The research uses publicly available datasets (CIFAR-10/100, STL-10, CelebA) and public pretrained models from HuggingFace. The work addresses IP protection, a legitimate research goal. No human subjects are involved.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: This work enables IP holders to verify model ownership with objective statistical evidence (positive impact). Potential risks include false ownership claims and retrospective selection of evidence sets; these are mitigated by requiring pre-registered watermarks with hash commitments, comparison against multiple public baselines, and principled significance thresholds (Section 4.4). Section 6 further discusses scope conditions.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: This paper proposes a verification protocol, not a new generative model or dataset. No new models or data with misuse risk are released.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All datasets (CIFAR-10/100, STL-10, CelebA) and pretrained models (HuggingFace DDPM checkpoints) are cited with their original publications in the references. These are standard research benchmarks with permissive research-use terms.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: The paper does not release new datasets, pretrained models, or code packages as assets.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: This work does not involve crowdsourcing or human subjects research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.

- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: This work does not involve human subjects research.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: LLMs are not used as a component of the core method. The method relies on diffusion model reconstruction error and Gaussian quantile regression, with no LLM involvement.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.